Applied Microbiology and Biotechnology (2026) 110:26
https://doi.org/10.1007/s00253-025-13703-y

RESEARCH

Enhancing the heat tolerance of homolactic *Escherichia coli* through reverse metabolic engineering

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Received: 15 July 2025 / Revised: 17 December 2025 / Accepted: 29 December 2025
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Abstract

Improving *Escherichia coli*'s thermotolerance through rational engineering is hindered by limited knowledge of the molecular mechanisms involved in supraoptimal thermal adaptation. To address this issue, we applied a reverse metabolic engineering strategy to develop a D-homolactic *E. coli* strain under non-aerated conditions. We first characterized the thermal reaction norms of the kinetic and stoichiometric parameters in strains evolved through continuous adaptive laboratory evolution, identifying those that maintained parental volumetric productivity and the lactate/glucose yield at high temperatures. Then, the genomic analysis of two thermally adapted strains was performed to determine the mutations acquired during thermal adaptation. Thermally adapted strains revealed convergent mutations in regulatory genes, notably *metJ*, *lrp*, and components of the RNA polymerase complex. Introducing these point mutations into the parental strain demonstrated that individual mutations in *rpoB* and *metJ* significantly improved growth at high temperatures, despite the presence of complex epistatic interactions in combinatorial analyses.

Key points

- *ALE significantly enhanced the thermotolerance of homolactic Escherichia coli*
- *Mutations in rpoB and metJ notably enhance growth at high temperatures*
- *Lactate productivity and yield stayed similar to the parental strain*

Keywords *Escherichia coli* · Thermotolerance · Reverse engineering · Lactic acid · *rpoB* · *metJ*

Introduction

Temperature is a critical abiotic factor that influences the survival of living organisms (Haynes 1964). In high-temperature bioprocesses, microbial thermotolerance is essential for producing biotechnologically valuable compounds (Matsushita et al. 2016; Mehta et al. 2016). Although thermophilic microorganisms capable of growing at elevated temperatures are available, their application is limited by complex growth requirements and the scarcity of molecular tools for metabolic engineering (Vavitsas et al. 2022).

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Published online: 17 January 2026

Escherichia coli remains a preferred platform for biotechnological production due to its well-characterized genetics (Yang et al. 2021), rapid growth, simple nutritional requirements, and a vast repertoire of molecular tools. However, its growth is restricted to mesophilic conditions (19 to 42 °C), with optimal growth occurring at 37 °C (Lenski and Bennett 1993), which limits its use in high-temperature processes.

Enhancing *E. coli* thermotolerance requires shifting its upper thermal limit. Reverse engineering, as outlined by Bailey et al. (2002), involves a three-step process: (i) defining a desired phenotype, (ii) identifying the genetic or environmental determinants of that phenotype, and (iii) transferring them through directed manipulation to endow that phenotype on another strain (Bailey et al. 2002). In this context, defining a thermotolerant *E. coli* phenotype requires stabilizing molecular and cellular processes at supraoptimal temperatures, balancing degeneration with the maintenance of critical functions (Caspeta et al. 2009).

Although short-term acclimatization through the heat shock stress response enables survival under transient stress, expanding *E. coli*'s thermal niche for industrial applications requires stable genetic modifications. Stable genetic changes are essential for allowing *E. coli* to thrive at high temperatures over extended periods. Understanding the molecular and cellular processes that govern thermotolerance, identifying the precise genetic components, their functions, and interactions that confer this trait presents a significant challenge. Direct engineering of thermotolerance is complex due to the incomplete understanding of the underlying molecular mechanisms. However, identifying barriers to growth at supraoptimal and lethal temperatures can guide the discovery of functional targets for genetic intervention.

In this study, we developed a reverse metabolic engineering approach to generate thermotolerant strains. To develop an *E. coli* strain for lactic acid production at high temperatures, we used a metabolically engineered D-homolactic strain for thermal adaptive laboratory evolution and to study in depth the effect of mutations acquired during the adaptation process. To achieve this, physiological analysis of thermally adapted strains enabled the identification of a phenotype capable of maintaining homolactic production at 47 °C in mineral medium supplemented with hydrolyzed protein. Comparative genomics of two adapted strains revealed convergent mutations in genes *rpoB*, *metJ*, and *lrp*, which were subsequently validated through genome editing. Single-point mutations in *rpoB* or *metJ* conferred thermotolerance to the parental strain at 45 °C; however, combinatorial analysis revealed only partial additive effects. These findings could help guide the rational design of robust, thermotolerant microbial cell factories for the production of other commodities. In addition, we expect that this information will contribute to understanding thermal adaptation at the molecular level. Finally, we anticipate that these findings will support future work aimed at improving tolerance to acids and inhibitory compounds, enabling optimal application in SSF processes.

Materials and methods

Organisms and culture conditions

The *E. coli* JU15 strain (MG1655: Δ*pflB*, Δ*frdA*, Δ*adhA*, Δ*xylFGH*, Δreg 27.3 kbp, gatC^{S184L}) (Utrilla et al. 2012, 2016) was used for the Adaptive Laboratory Evolution (ALE) experiments. Parental, intermediate, and endpoint strains were evaluated in batch fermentations using AM1 minimal medium supplemented with minimal concentrations of tryptone and yeast extract. The AM1 minimal medium (Martinez et al. 2007) was composed of 2.63 g/L $(NH_4)_2HPO_4$, 0.87 g/L $NH_4H_2PO_4$, 1.5 mM $MgSO_4{\cdot}7H_2O$, 1.5 mL/L trace metal solution, 2.0 mM KCl, and 1.0 mM betaine. The trace metal solution contains per liter: 1.6 g $FeCl_3$, 0.2 g $CoCl_2{\cdot}6H_2O$, 0.1 g $CuCl_2$, 0.2 g $ZnCl_2{\cdot}4H_2O$, 0.2 g Na_2MoO_4, 0.05 g H_3BO_3, and 0.33 g $MnCl_2{\cdot}4H_2O$. The AM1 medium was supplemented with 0.1 g/L of sodium citrate, 40 g/L of glucose, and 0.1 or 1 g/L of tryptone along with 0.05 or 0.5 g/L of yeast extract. The final concentrations of tryptone and yeast extract corresponded to 1% and 10% of the Luria Broth (LB) medium composition, respectively, without NaCl.

Thermal experimental evolution

Experimental evolution was performed using the *E. coli* ECL45 strain, following the methodology previously described by Pérez-Morales et al. (2024). Briefly, the initial inoculum, with an optical density at 600 nm (OD_{600}) of 0.1, was prepared from a liquid culture derived from the largest colony on a Petri dish. The chemostat culture was carried out using a 1-L bioreactor (Applikon Ez-control, Delft, NL) with a working volume of 750 mL. Agitation was maintained at 400 rpm, and the pH was automatically controlled at neutrality by the addition of an 8 N KOH solution. The culture was initiated in batch mode at 45 °C without aeration, starting from an OD_{600} of 0.1 (equivalent to 0.037 g_{DCW}/L). After reaching the stationary phase (approximately 24 h), continuous culture was established at a flow rate of 1.125 mL/min of concentrated medium, corresponding to a dilution rate (D) of 0.09 h^{-1}. During evolution, the temperature was increased progressively by 0.1–0.2 °C after every three residence times at steady state. Cell concentrations were maintained within the initial range throughout the experiment.

The culture medium used for chemostat operation was AM1 minimal medium supplemented with 40 g/L glucose, 0.1 g/L sodium citrate, 0.1 or 1 g/L tryptone, and 0.05 or

0.5 g/L yeast extract. Before each temperature increment, 0.8 mL samples were collected, mixed with 40% (v/v) glycerol, and stored at − 70 °C. Monoclonal thermotolerant strains were subsequently isolated and evaluated from colonies obtained from the cryopreserved samples.

Physiological analysis

The physiological measurements were obtained from batch cultures grown in AM1 minimal medium supplemented with 40 g/L of glucose and 0.1 g/L of sodium citrate, using 300-mL mini-fermenters (Beall et al. 1991). Each mini-fermenter contained 200 mL of culture medium and operated at 150 rpm without aeration. Cultures were inoculated at an initial OD_{600} of 0.1. The temperature was maintained using a thermocirculator, and the pH was controlled automatically by adding 2 N KOH. Tryptone (0.1 or 1 g/L) and yeast extract (0.05 or 0.5 g/L) were supplemented at concentrations equivalent to 1% and 10% of Luria Broth (LB) medium composition without NaCl, respectively. Kinetic and stoichiometric parameters were derived from growth curves, glucose consumption profiles, and D-lactic production measurements.

Genomic analysis

Genomic DNA was extracted from liquid cultures in LB medium using the DNeasy Ultraclean Microbial Kit (QIAGEN, Hilden, Germany). Whole-genome sequencing was performed on the Illumina® NextSeq 500 platform, generating paired-end reads of 2 × 75 bp. Genome assemblies were conducted using BWA-MEM (Li and Durbin 2009) by aligning paired reads to the *E. coli* K12 MG1655 reference genome (NCBI access: NC_000913.3). Read quality control and preprocessing were performed using FastQC (Andrews 2010) and Cutadapt (Martin 2011). SAM/BAM files were processed with Samtools (Li et al. 2009). Genome annotation was conducted using Prokka (Seemann 2014), with reference to the *E. coli* K12 MG1655 genome. Variant identification and mapping of mutations in thermotolerant strains were performed using the Breseq pipeline (Deatherage and Barrick 2014) relative to the *E. coli* JU15 reference genome. The sequencing raw data was deposited in the NCBI as BioProject PRJNA1295688 (https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1295688).

Genome editing

We implemented a reverse engineering strategy to introduce selected mutations from the evolved GT48 strain into the parental *E. coli* JU15 background, assessing the functional impact of specific mutations acquired during thermal adaptation. The targeted mutations were located in regulatory genes associated with gene expression, specifically *rpoB*, *metJ*, and *lrp*. Genome editing was performed using a two-step CRISPR-Cas9 system, as previously described (Jiang et al. 2015; Wang et al. 2018). Briefly, in the first step, an artificial 20-bp guide sequence (N20) was inserted at the target genomic locus. This artificial site was then used in the second step to direct the introduction of the desired point mutation. To implement this approach, electrocompetent cells were prepared, pTargets plasmids were constructed (Table 1), and donor DNA sequences were amplified.

Electrocompetent cells were prepared using the method described by Miller and Nickoloff (1995). Plasmids pCas and pTarget were obtained from Addgene (http://www.addgene.org/crispr-cas) (Table 1). To modify the guide sequence in pTarget, the plasmid was linearized with LV1 and LV2 primers (Reyes-González et al. 2021) and recircularized using complementary oligonucleotides (Table 2) via the circular polymerase extension cloning (CPEC) method (Quan and Tian 2011). Donor sequences containing the artificial N20 sequence were generated by overlapping PCR of two previously amplified fragments (primers listed in Table 2) and purified using the GeneJET PCR Purification Kit (Thermo Scientific, Waltham, MA, USA). Final donor templates carrying the desired point mutations were amplified from the GT48 genome, using the primers described in Table 2. The amplified fragments were resolved on a 0.8% (w/w) agarose gel and purified for downstream use.

The first round of CRISPR/Cas9 editing involved a two-step transformation. First, 100 ng of the pCas plasmid was introduced into electrocompetent target cells. Subsequently, these pCas-positive cells were transformed with 400 ng of the donor DNA (D1) and 100 ng of the modified pTarget-T1 plasmid. Transformations were performed using standard laboratory protocols. Briefly, after electroporation, cells were recovered in 950 µL of SOC medium and incubated at 30 °C with shaking at 300 rpm for three h. Subsequently, cells were plated on LB agar containing 50 µg/mL kanamycin and 50 µg/mL spectinomycin, then incubated overnight at 30 °C. Candidate colonies were screened by colony PCR. The pTarget plasmid was cured by inoculating positive colonies in 2 mL of LB with 50 µg/mL kanamycin and 0.5 mM IPTG, followed by plating on LB-kanamycin plates. The

Table 1 Plasmids used in this study

Plasmid	Source or reference
pCas	Addgene: 62225
pTarget	Addgene:62226
pTarget-N20	This study
pTargetT1-rpoB	This study
PTarget-T1-metJ	This study
pTarget-T1-lrp	This study

Table 2 Primers used for CRISPR-Cas9 genome editing

Name	Sequence (5′ 3′)
N20 artificial	cagctctcggcactgtaatc
LV1	ctcgagcgatcgaggaaagaacatgtgagc
LV2	gttttagagctagaaatagcaag
pT2N20Fw	ggtataatactagtcagctctcggcactgtaatcgttttagagcta-gaaatagcaag
pT2N20Re	agctgactagtattatacctaggactgagctagctgtcaactcgagc-gatcgag
pT1RpbFw	ggtataatactagtaggtatgccaatcgcaacgcgttttagagcta-gaaatagcaag
pT1RpbRe	tacctactagtattatacctaggactgagctagctgtcaactcgagcgatc-gag
RpbFwD1	gctgaccccagaagaaaaactgc
RpbReD2	cgtcggagttagcacaatccgc
RpbReD4	gattacagtgccgagagctgttgcgcaggttttcagccag
RpbFwD3	cagctctcggcactgtaatccggtgttcgacggtgcgaaagaag
pT1MetFw	ggtataatactagtttacgcagatcggcatcatcgttttagagcta-gaaatagcaag
pT1MetRe	cgtaaactagtattatacctaggactgagctagctgtcaactcgagc-gatcgag
MetFwD1	gttgttcaacactcagtccgc
MetReD2	gttcgtcgtcatttaacccgc
MetReD4	gattacagtgccgagagctgagagcgcagcgacgaaatc
MetFwD3	cagctctcggcactgtaatccggcaaaggttgcccggtaa
pT1LrpFw	ggtataatactagtgtccgtgtgtcattgacgccgttttagagcta-gaaatagcaag
pT1LrpRe	cggacactagtattatacctaggactgagctagctgtcaactcgagc-gatcgag
LrpFwD1	gccattcgtgatgtcgatagc
LrpReD2	gtgtagcagtggttgtagcg
LrpReD4	cagctctcggcactgtaatcaggcagacgcagcagggt
LrpFwD3	gattacagtgccgagagctgatacgttgttatggaagaagtcaagc

absence of growth on LB-spectinomycin plates confirmed plasmid curing. The second editing step followed the same procedure. In this round, the pTarget-N20 plasmid carried a guide sequence targeting the artificial N20 site, while the donor templates (D2) encoded specific point mutations. After transformation, colonies were screened by PCR and confirmed through Sanger sequencing. The curing of pTarget was repeated as described above. The final curing of the pCas plasmid was achieved by incubating the strains overnight at 43 °C in LB medium without antibiotics. Colonies were tested for plasmid loss by streaking on LB agar plates with and without kanamycin and spectinomycin.

Analytical methods

Cellular concentration was determined indirectly by measuring the optical density at 600 nm (OD_{600}) using a Genesys 10S UV–VIS spectrophotometer (Thermo Scientific, Waltham, MA, USA). OD_{600} values were converted to dry cell weight (DCW) using the conversion factor of 1.0 $OD_{600} = 0.37\ g_{DCW}/L$. After the OD_{600} measurement, culture samples were immediately centrifuged, and the supernatant was stored at − 4 °C for subsequent analysis. Glucose and D-lactic acid in the supernatant were quantified by High-Performance Liquid Chromatography (HPLC) using a Waters system (Waters, MA, USA). Metabolite separation was performed on an Aminex HPX-87H column (Bio-Rad, Hercules, CA, USA) maintained at 60 °C. The mobile phase consisted of 5 mM H_2SO_4, delivered at a flow rate of 0.5 mL/min. Detection was carried out using a diode array detector (Waters 996, MA, USA) and a refractive index detector (Waters 410, MA, USA). Metabolite concentrations were calculated based on calibration curves generated from analytical grade standards.

Results

Fitness recovery through continuous adaptive laboratory evolution

Several strategies have been developed to enhance thermotolerance in *Escherichia coli*. However, most approaches confer only transient heat resistance, which limits their applicability in long-term high-temperature bioprocesses (Pérez-Morales et al. 2025). Adaptive Laboratory Evolution (ALE) has been shown to be an effective method for selecting adapted *E. coli* strains (Sandberg et al. 2019) (Fig. 1). Nevertheless, previous ALE studies have primarily involved wild-type strains with unaltered redox balance and central metabolism (Blaby et al. 2012; Kishimoto et al. 2010; Rudolph et al. 2010; Sandberg et al. 2014; Tenaillon et al. 2012).

Thermal reaction norms of kinetic parameters

During the ALE experiment, individual colonies were periodically isolated and evaluated in batch fermentations at the temperature corresponding to the evolutionary stage at which they were collected. Colonies exhibiting higher growth rates and volumetric productivities were selected for further characterization across a range of temperatures. From these batch cultures, the thermal reaction norms of key kinetic and stoichiometric parameters were determined. As shown in Fig. 2, the parental *E. coli* strain JU15 demonstrated the capacity to adapt to temperatures up to 46.5 °C when cultivated in glucose-minimal medium supplemented with only 1% LB components. The reaction norms of the kinetic and stoichiometric parameters revealed a restoration of fitness at elevated temperatures, with no observable trade-offs relative to the ancestral performance under optimal conditions.

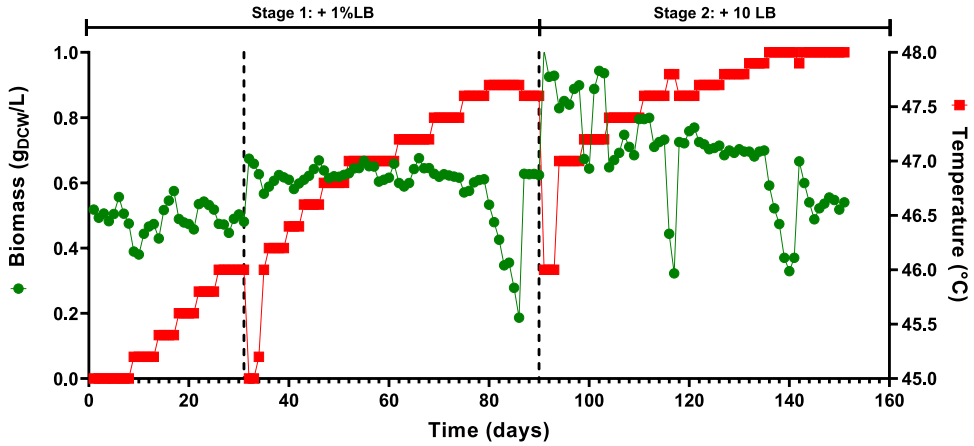


Fig. 1 Fitness growth recovery during the thermal experimental evolution in chemostat culture

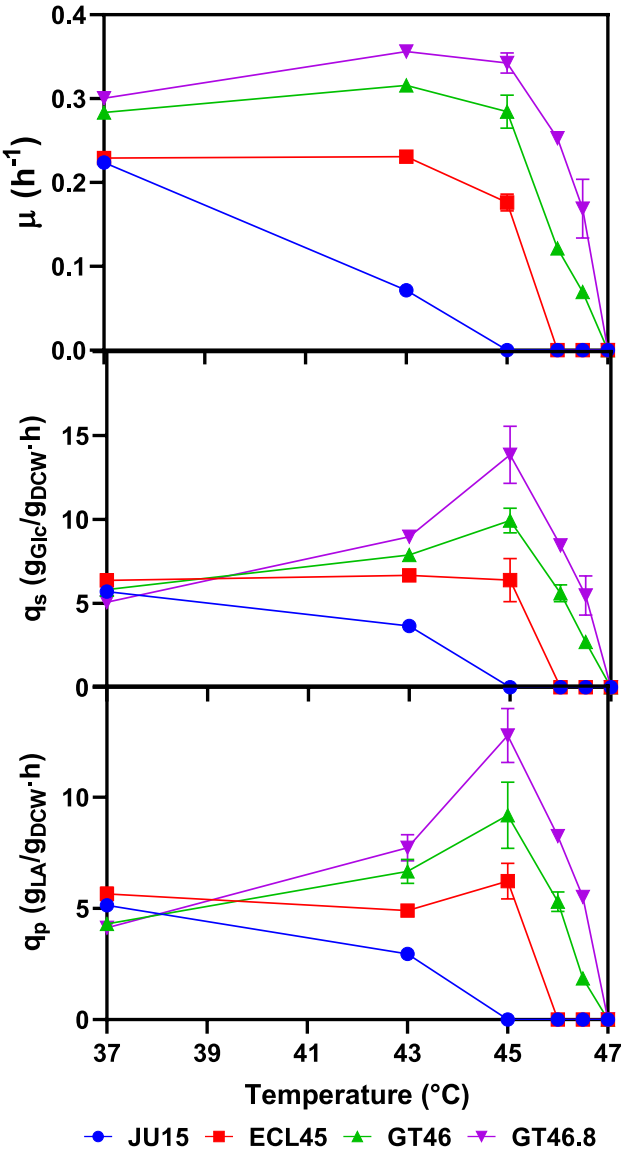


Fig. 2 Thermal reaction norms, with 0.15 g/L hydrolyzed protein, of the specific growth rate (μ), the specific glucose consumption rate (q_S), and the specific lactate production rate (q_P)

In terms of growth rate, the JU15 parental strain could grow at temperatures up to 43 °C. The ECL45 strain, isolated at the endpoint of the ALE by batch–serial passages, extended this upper limit to 45 °C. Notably, the GT46.8 strain, isolated before the endpoint of Stage 1 in chemostat ALE, exhibited superior growth rates compared to the parental and intermediate strains derived during the ALE process. Although biomass accumulation at 24 h and the biomass-to-substrate yield decreased with increasing temperature across all thermally adapted strains, both volumetric productivity and product-to-substrate yield remained comparable to those of the parental strain. Additionally, the specific glucose consumption rate and the D-lactic acid production rate peaked at 45 °C, indicating a thermal optimum for metabolic activity. Interestingly, strains isolated during the final stage of ALE exhibited a marked increase in both glucose consumption and lactic acid production rates at 45 °C, suggesting enhanced metabolic efficiency near the upper thermal limit.

On the other hand, under conditions supplemented with 10% LB medium, the thermal reaction norms revealed that while the parental strain JU15 is capable of growth up to 45 °C, the evolved strain demonstrated sustained growth at temperatures up to 47 °C (Fig. 3). The thermally adapted strain exhibited consistent trends in its kinetic and stoichiometric parameters across the entire temperature range evaluated. Notably, volumetric productivity and biomass accumulation at 24 h remained comparable to those of the parental strain. While the product-to-substrate yield increased at higher temperatures, the biomass-to-substrate yield decreased, indicating a metabolic shift toward product formation over biomass synthesis. Strikingly, the thermally adapted strain displayed peak values of specific glucose consumption and D-lactate production rates at 46.5 °C, indicating optimal activation of central metabolic processes at this temperature.

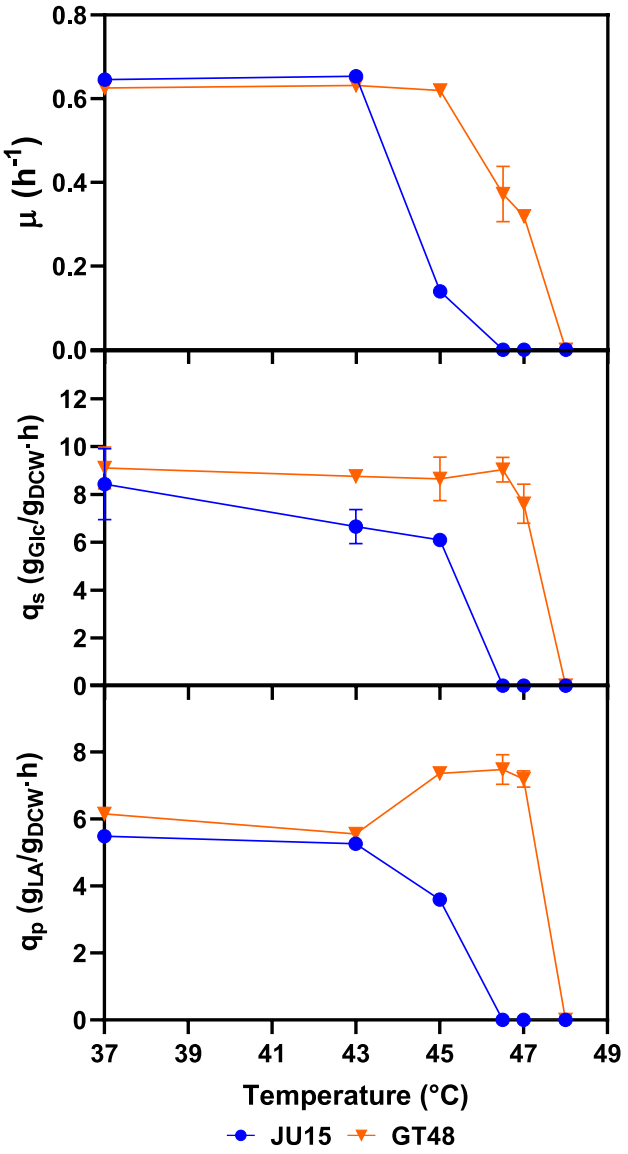


Fig. 3 Thermal reaction norms, with 1.5 g/L hydrolyzed protein, of the specific growth rate (μ), the specific glucose consumption rate (q_S), and the specific lactate production rate (q_P)

Mutational changes in regulatory genes converge during thermal adaptation

To identify potential targets for improving thermotolerance through reverse engineering, we performed a comparative genomic analysis of two thermally adapted *E. coli* strains (Table 3). The strain GT46.8, isolated at the endpoint of Stage 1 with a maximum adaptation temperature of 46.5 °C, exhibited 10 mutations: 5 non-synonymous, 2 synonymous, 2 intergenic, and 1 single-base-pair insertion. In contrast, the strain GT48, obtained at the endpoint of Stage 2 with growth sustained at 47 °C, accumulated 15 mutations: 5 non-synonymous, 2 synonymous, 6 intergenic, and 2 located in non-coding regions. Notably, the eight mutations acquired during the first phase of evolution in the JU15 strain were retained in both evolved strains. In total, 23 new mutations were identified across various functional categories, including metabolism, gene regulation, transcriptional machinery, cell envelope, and translational machinery, as well as intergenic mutations.

The mutational landscape observed reflects the specificity of the ALE conditions, wherein identical mutations across replicates are rare. However, adaptive evolution often exhibits convergence at higher levels of biological organization, namely at the gene, operon, or protein complex level (Tenaillon et al. 2012). In this study, both thermally adapted strains showed convergent mutational patterns at the site-specific, gene, and functional complex levels, particularly in genes involved in global regulatory and gene expression functions (Fig. 4). These results suggest a common adaptive strategy for coping with high-temperature stress in the homolactic *E. coli* background.

Reverse engineering of regulatory genes improves thermal tolerance

Previous studies have suggested that early mutational events during thermal adaptation are primarily directed toward restoring gene expression as a compensatory mechanism to maintain homeostasis under heat stress (Rodriguez-Verdugo et al., 2016; Ying et al. 2015). In this context, we hypothesized that the targeted introduction of specific mutations in key regulatory genes could reproduce a thermotolerant phenotype through reverse metabolic engineering. To test this hypothesis, CRISPR/Cas9-mediated genome editing was used to introduce convergent mutations identified in the GT48 endpoint strain ($MetJ^{D76A}$, $RpoB^{T1184A}$, and Lrp^{V141I}) into the genetic background of the parental strain JU15. Physiological characterization of these engineered strains revealed that individual point mutations in *rpoB* and *metJ* conferred increased thermotolerance at 45 °C in 10% LB, improving fitness growth by 37% and 27%, respectively, compared to the unmodified parental strain (Fig. 5). In contrast, the *lrp* mutation alone did not significantly affect thermotolerance.

Combinatorial analysis of mutations revealed complex epistatic interactions. Specifically, the double mutants *rpoB-lrp* (RBL strain) and *metJ-lrp* (MJL strain) exhibited enhanced thermotolerance, with improvements of 53% and 28%, respectively. However, the combination of *rpoB* and *metJ* mutants (HM2 strain) abolished the thermotolerant phenotype observed in the individual mutants, indicating a case of diminishing returns epistasis. Moreover, the triple

Table 3 Mutational analysis between parental and thermally adapted strains

Mutation	Annotation	Gene	ECL45	GT46.8	GT48
A → C	S159R (AGT → CGT)	*dxr* →	X	X	X
C → G	G235A (GGT → GCT)	*sapD* ←	X	X	X
$(A)_{6 \rightarrow 5}$	Intergenic (−37/+325)	*gadB* ←/← *pqqL*	X	X	X
G → T	P460T (CCA → ACA)	*cysJ* ←	X	X	X
Δ1 bp	coding (258/564 nt)	*yqgE* →	X	X	X
C → G	L277V (CTG → GTG)	*spoT* →	X	X	X
C → T	A456V (GCT → GTT)	*spoT* →	X	X	X
G → T	D248E (GAC → GAA)	*argE* ←	X	X	X
C → A	A7A (GCC → GCA)	*ilvI* →			X
G → A	D114N (GAC → AAC)	*lrp* →		X	
G → A	V141I (GTC → ATC)	*lrp* →			X
$(A)_{6 \rightarrow 5}$	intergenic (−192/+411)	*ompF* ←/← *asnS*		X	X
C → T	Intergenic (+262/+1040)	*tdk* →/← *adhE*			X
G → T	T38N (ACC → AAC)	*ldhA* ←		X	
C → A	intergenic (−36/+29)	*yegX* ←/← *thiD*			X
G → A	L117L (CTG → TTG)	*mepA* ←		X	
$(TATGGCAC)_{6 \rightarrow 7}$	intergenic (−310/+50)	*abpA* ←/← *yfjM*			X
C → A	intergenic (+262/−92)	*nrdF* →/→ *proV*		X	
$(T)_{8 \rightarrow 9}$	intergenic (+2159/+357)	*ygeI* →/← *insD4*			X
A → G	L254P (CTG → CCG)	*rpoA* ←		X	
C → A	R120L (CGC → CTC)	*yheO* ←		X	
A → G	*320* (TAA → TAG)	*waaC* →		X	
A → T	intergenic (+161/−23)	*alaT* →/→ *rrlA*			X
C → G	noncoding (1868/2905 nt)	*rrlA* →			X
G → C	noncoding (1873/2905 nt)	*rrlA* →			X
T → G	D76A (GAT → GCT)	*metJ* ←		X	X
A → G	T1184A (ACG → GCG)	*rpoB* →			X
G → T	G165C (GGC → TGC)	*prop* →			X
$(T)_{5 \rightarrow 4}$	coding (13/735 nt)	*frdB* ←		X	
T → C	F35F (TTT → TTC)	*fimA* →			X
G → A	E68K (GAA → AAA)	*fimA* →			X

mutations in ECL45 were previously reported by Pérez-Morales et al. (2024)

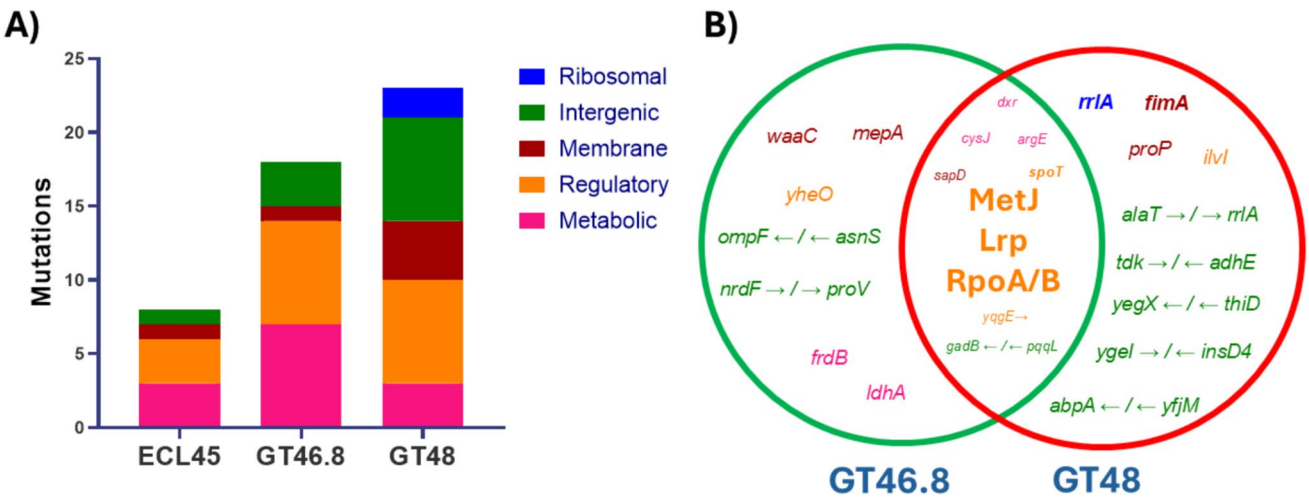


Fig. 4 Mutational changes acquired during thermal evolution. **A** Functional classification of mutations present in thermally adapted strains. **B** Convergent and shared mutations present in thermally adapted strains

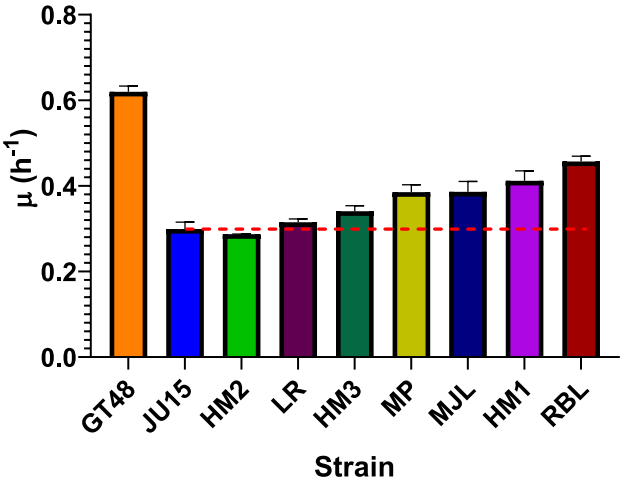


Fig. 5 Mutational effect on fitness growth at 45 °C. Cultivations were performed in the AM1 mineral medium with 40 g/L glucose and 1.5 g/L hydrolyzed protein; HM1 (*RpoB*^{T1184A}), MP (*MetJ*^{D76A}), HM2 (*RpoB*^{T1184A}, *MetJ*^{D76A}), LR (*Lrp*^{V141I}), RBL (*RpoB*^{T1184A}, *Lrp*^{V141I}), MJL (*MetJ*^{D76A}, *Lrp*^{V141I}), HM3 (*MetJ*^{D76A}, *RpoB*^{T1184A}, *Lrp*.^{V141I})

mutant HM3 strain (*rpoB-metJ-lrp*) did not exhibit any additional benefit over the parental JU15 strain, further supporting the existence of non-additive epistatic interactions between these regulatory loci.

Given that epistatic effects can be context-dependent, additional tests were conducted in AM1 mineral medium supplemented with 1% LB components at 45 °C to assess the stability of the thermotolerant phenotype under nutrient-limited conditions (Fig. 6). Under these conditions, strains harboring single mutations in either *rpoB* or *metJ* maintained their enhanced thermotolerance, confirming their robustness across various environmental settings.

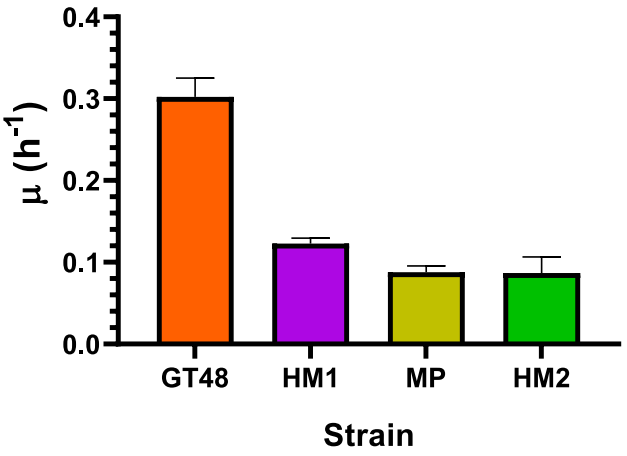


Fig. 6 Specific growth rate at 45 °C in AM1 mineral medium with 0.15 g/L hydrolyzed protein; HM1 (*RpoB*^{T1184A}), MP (*MetJ*^{D76A}), HM2 (*RpoB*^{T1184A}, *MetJ*^{D76A})

Discussion

In this study, we used ALE on a metabolically engineered homolactic *E. coli* strain (JU15) to enhance thermotolerance under specific conditions: glucose-minimal medium, pH 7, and non-aerated cultivation. The previously evolved strain ECL45, capable of growth at 45 °C (Pérez-Morales et al. 2025), was subjected to a second round of ALE under continuous culture conditions (chemostat) with progressive temperature increases above 45 °C. Continuous cultivation was selected over serial batch transfers to avoid bottleneck effects and promote gradual adaptation. The endpoint strain was successfully adapted to 48 °C through a multi-phase ALE process that included modifications to the bioreactor configuration and medium composition. During chemostat cultivation, the temperature was incrementally raised, and adaptation was confirmed by the recovery of steady-state growth rates at each step.

Thermal reaction norms of kinetic parameters suggest that thermally evolved cells exhibit enhanced enzymatic activity in pathways related to glycolysis and biosynthetic processes in response to heat stress. This observation aligns with predictions from genome-scale metabolic models, which propose that adaptation to elevated temperatures, such as 45 °C, may involve increased catalytic activity of key enzymes required for cellular component synthesis (Chang et al. 2013; Chen et al. 2017). Interestingly, similar to previously characterized thermotolerant wild-type strains (Sandberg et al. 2014), the thermotolerant homolactic strain in this study demonstrated concurrent improvements in growth rate, glucose uptake, and metabolite production at supraoptimal temperatures. These results support the notion that both wild-type and engineered *E. coli* strains can adapt to high-temperature environments through convergent physiological strategies that enhance metabolic throughput.

In prior ALE experiments using serial batch transfers, supplementation with LB-rich medium enabled JU15 to grow at temperatures up to 45 °C (Pérez-Morales et al. 2025). Elevated temperatures (> 42 °C) are known to impair *E. coli* growth due to thermosensitive limitations in the synthesis of essential cofactors and amino acids (Chang et al. 2013; Mordukhova et al. 2008; Mordukhova and Pan 2013). While thermotolerant phenotypes have been achieved in wild-type strains at temperatures up to 48 °C using LB-rich media (Blaby et al. 2012; Rudolph et al. 2010), the cost and complexity of rich media limit their industrial applicability. To address this issue, the concentration of LB components (tryptone and yeast extract) was capped at 10% of standard LB medium. During early ALE stages, supplementation with only 1% LB enabled thermal adaptation up to 46.5 °C (Stage 1). In Stage 2, the concentration was increased to 10%, which supported adaptation to 48 °C under continuous cultivation

conditions (Fig. 1). These results underscore the importance of both controlled environmental parameters and strategic nutrient supplementation in guiding the thermal adaptation of metabolically engineered strains.

Among the most prominent mutational changes that converge during thermal adaptation in regulatory genes were those in the genes encoding the global regulators *lrp* and *metJ*, as well as genes encoding subunits of the RNA polymerase complex, including *rpoA* and *rpoB*. The RNA polymerase holoenzyme is central to transcription regulation, and mutations in *rpoB* have previously been implicated in modulating global gene expression to restore homeostasis under thermal stress (González-González et al. 2017; Rodríguez-Verdugo et al. 2016). While RNA polymerase mutations are often detected during the early stages of thermal adaptation, the presence of mutations in *rpoB* and *rpoA* at later ALE stages in this study suggests a secondary role, possibly in facilitating transcriptional stability at temperatures above 45 °C. This divergence in timing and function may reflect differences in genetic background or cultivation conditions. Since the substitution at position Lrp^{D114N} has already been explored in previous works, we focused on the Lrp^{V141I} variant. Based on this, we would expect the Lrp^{D114N} variant to have a similar effect, as it affects the same protein domain and is predicted to disrupt the standard leucine-dependent control of Lrp activity even more strongly than the previously evaluated Lrp^{D114E} variant (Ziegler & Freddolino 2023).

Unlike previous ALE studies involving wild-type *E. coli*, which reported a broader range of mutation types, including duplications, deletions, insertions, and point mutations (Luan et al. 2015; Riehle et al. 2001), our analysis identified only single-nucleotide polymorphisms (SNPs) and insertions or deletions (indels), located within coding, non-coding, and intergenic regions. The increased proportion of synonymous and intergenic mutations in the GT48 strain may reflect a reduced availability of beneficial mutations as thermotolerance nears its upper physiological limit. This shift from adaptive to neutral mutations has been observed previously during ALE under minimal medium conditions. It may indicate a need for extended evolutionary time or elevated mutation rates to reintroduce beneficial variation. Although no mutations were detected in DNA repair genes in this study, other research suggests that thermal stress may promote hypermutator phenotypes (Kang et al. 2019; Kishimoto et al. 2010; Sandberg et al. 2014). In support of this, the use of mutagenic plasmids has been shown to enhance mutation rates and enable ALE-based thermotolerance increases in *E. coli* to temperatures up to 50 °C (Luan et al. 2015).

This study successfully applied reverse metabolic engineering to generate a thermotolerant phenotype in a D-homolactic *E. coli* strain by leveraging physiological and genomic data from strains evolved through continuous adaptive laboratory evolution. It provides fundamental information on potential thermotolerance-related targets for reverse engineering. We identified regulatory gene targets associated with enhanced thermotolerance. A single point mutation in either *rpoB* or metJ was sufficient to confer growth capacity at 45 °C without compromising key physiological traits such as biomass yield and product formation.

Although our strategy focused on selecting a set of convergent regulatory genes during adaptation, further investigations are required to elucidate the effects of other mutations on thermotolerance. Other regulatory genes, such as *spoT*, which participates in stress-mediated responses (Potrykus and Cashel 2008), and hypothetical genes such as *yheO* (Duarte-Velázquez et al. 2022) and *yqgE* (Al Mamun et al. 2012), which have been suggested to play a possible role in the transcriptomic genetic network, could be targets to restore a homeostatic transcriptomic response at high temperatures. In this context, mutations in intergenic regions should be examined to determine how regulatory elements may be altered.

Consistent with structural genome-scale metabolic models, which suggest that protein structural properties determine metabolic thermotolerance, we expect that some mutations in metabolic proteins may be beneficial by increasing the thermal stability of their activity and thereby maintaining metabolic flux (Chang et al. 2013). In this context, mutations in *ldh* are a promising target for evaluation, as increased lactic acid production in *E. coli* has been reported under high-temperature exposure (Kim et al. 2020).

Membrane-related proteins have previously been explored as targets to increase thermotolerance, and these studies have shown that mutations can affect membrane composition. In our case, membrane composition is unlikely to be altered because the mutations in *mepA* and *waaC* are synonymous. Instead, we would only expect potential changes in the function of membrane transporters, such as *proP*-mediated osmoregulation, given that the medium contains betaine as an osmoprotectant (MacMillan et al. 1999). Non-coding ribosomal RNA sequences show a positive relationship with temperature. In particular, the correlation between the G + C content of rRNA stem regions and optimal growth temperature has been used to infer the optimal growth temperature of ancestral organisms (Morales and Delaye 2020). Within this framework, one might expect that mutational changes increasing G + C content would enhance thermotolerance. However, in the strain GT48, the observed mutations do not alter the G + C content, either in its distribution along the molecule or in its overall value, and thermotolerance therefore remains unchanged.

Our findings underscore the utility of reverse metabolic engineering as a rational strategy for improving microbial thermotolerance. Furthermore, the observed epistatic interactions highlight the importance of combinatorial and context-specific analyses when designing multi-gene modifications. Overall,

this work establishes a framework for improving the thermal performance of *E. coli* engineered strains through targeted and evolution-informed genetic interventions. It is worth noting that acid tolerance is another important environmental factor to consider when developing *E. coli* strains for commercial-scale lactic acid production. Hence, future work will focus on developing *E. coli* strains, from the thermotolerant strain GT48, for lactate production at acidic pH.

Acknowledgements We thank Fernando Astudillo, Ignacio Ramírez, Omar Cortés, Angelica Vallejo, and Nadia Varela for their critical comments and some technical support. We thank Eliseo Molina-Vázquez, Georgina Hernández-Chávez, Luz María Martínez-Mejía, Noemí Flores-Mejía, Martín Patiño-Vera, and Mario Trejo-Loyo for their technical assistance.

Author contribution GPM: study conceptualization, methodology, experimentation, data analysis, writing—original draft, and writing—review; JVC: experimentation; EM: data analysis and writing—review and editing; MAC: data analysis and writing—review and editing; GG: data analysis and writing—review and editing; AM: study conceptualization, methodology, data analysis, supervision, funding acquisition, project administration, writing—original draft, and writing—review and editing

Funding This work was supported by the Secretariat of Science, Humanities, Technology, and Innovation - Mexico (SECIHTI-México) grant CF-2023-I-1116. GPM held a scholarship from SECIHTI.

Data availability Data will be made available upon request.

Declarations

Ethics approval and consent to participate This article contains no studies with human participants or animals performed by any of the authors.

Competing interests The authors declare no competing interests.

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References

Al Mamun AAM, Lombardo MJ, Shee C, Lisewski AM, Gonzalez C, Lin D, Nehring RB, Saint-Ruf C, Gibson JL, Frisch RL, Lichtarge O, Hastings PJ, Rosenberg SM (2012) Identity and function of a large gene network underlying mutagenic repair of DNA breaks. Science 338(6112):1344–1348. https://doi.org/10.1126/science.1226683

Andrews S (2010) FastQC - A quality control tool for high throughput sequence data. Babraham Bioinform. http://www.bioinformatics.babraham.ac.uk/projects/fastqc/. Accessed 15 Jan 2025

Bailey JE, Sburlati A, Hatzimanikatis V, Lee K, Renner WA, Tsai PS (2002) Inverse metabolic engineering: a strategy for directed genetic engineering of useful phenotypes. Biotechnol Bioeng 79:109–121. https://doi.org/10.1002/bit.10441

Beall DS, Ohta K, Ingram LO (1991) Parametric studies of ethanol production form xylose and other sugars by recombinant *Escherichia coli*. Biotechnol Bioeng 38:296–303. https://doi.org/10.1002/bit.260380311

Blaby IK, Lyons BJ, Wroclawska-Hughes E, Phillips GCF, Pyle TP, Chamberlin SG, Benner SA, Lyons TJ, de Crécy-Lagard V, de Crécy E (2012) Experimental evolution of a facultative thermophile from a mesophilic ancestor. Appl Environ Microbiol 78:144–155. https://doi.org/10.1128/AEM.05773-11

Caspeta L, Flores N, Pérez NO, Bolívar F, Ramírez OT (2009) The effect of heating rate on *Escherichia coli* metabolism, physiological stress, transcriptional response, and production of temperature-induced recombinant protein: a scale-down study. Biotechnol Bioeng 102:468–482. https://doi.org/10.1002/bit.22084

Chang RL, Andrews K, Kim D, Li Z, Godzik A, Palsson BO (2013) Structural systems biology evaluation of metabolic thermotolerance in *Escherichia coli*. Science (1979) 340:1220–1223. https://doi.org/10.1126/science.1234012

Chen K, Gao Y, Mih N, O'Brien EJ, Yang L, Palsson BO (2017) Thermosensitivity of growth is determined by chaperone-mediated proteome reallocation. Proc Natl Acad Sci USA 114(43):11548–11553. https://doi.org/10.1073/pnas.1705524114

Deatherage DE, Barrick JE (2014) Identification of mutations in laboratory-evolved microbes from next-generation sequencing data using breseq. Methods Mol Biol 1151:165–188. https://doi.org/10.1007/978-1-4939-0554-6_12

Duarte-Velázquez I, de la Mora J, Ramírez-Prado JH, Aguillón-Bárcenas A, Tornero-Gutiérrez F, Cordero-Loreto E, Anaya-Velázquez F, Páramo-Pérez I, Rangel-Serrano Á, Muñoz-Carranza SR, Romero-González OE, Cardoso-Reyes LR, Rodríguez-Ojeda RA, Mora-Montes HM, Vargas-Maya NI, Padilla-Vaca F, Franco B (2022) *Escherichia coli* transcription factors of unknown function: sequence features and possible evolutionary relationships. PeerJ 10:e13772. https://doi.org/10.7717/peerj.13772

González-González A, Hug SM, Rodríguez-Verdugo A, Patel JS, Gaut BS (2017) Adaptive mutations in RNA polymerase and the transcriptional terminator *rho* have similar effects on *Escherichia coli* gene expression. Mol Biol Evol 34(11):2839–2855. https://doi.org/10.1093/molbev/msx216

Haynes TM (1964) The effects of temperature on living organisms. Am Biol Teach 26(7):511–514. https://doi.org/10.2307/4440735

Jiang Y, Chen B, Duan C, Sun B, Yang J, Yang S (2015) Multigene editing in the *Escherichia coli* genome via the CRISPR-Cas9 system. Appl Environ Microbiol 81(7):2506–2514. https://doi.org/10.1128/AEM.04023-14

Kang M, Kim K, Choe D, Cho S, Kim SC, Palsson B, Cho BK (2019) Inactivation of a mismatch-repair system diversifies genotypic landscape of *Escherichia coli* during adaptive laboratory evolution. Front Microbiol 10:1845. https://doi.org/10.3389/fmicb.2019.01845

Kim S, Kim Y, Suh DH, Lee CH, Yoo SM, Lee SY, Yoon SH (2020) Heat-responsive and time-resolved transcriptome and metabolome analyses of *Escherichia coli* uncover thermo-tolerant mechanisms. Sci Rep 10(1):17715. https://doi.org/10.1038/s41598-020-74606-8

Kishimoto T, Iijima L, Tatsumi M, Ono N, Oyake A, Hashimoto T, Matsuo M, Okubo M, Suzuki S, Mori K, Kashiwagi A, Furusawa C, Ying BW, Yomo T (2010) Transition from positive to neutral in mutation fixation along with continuing rising fitness in thermal

adaptive evolution. PLoS Genet 6(10):e1001164. https://doi.org/10.1371/journal.pgen.1001164
Lenski RE, Bennett AF (1993) Evolutionary response of *Escherichia coli* to thermal stress. Am Nat. https://doi.org/10.1086/285522
Li H, Durbin R (2009) Fast and accurate short read alignment with Burrows-Wheeler transform. Bioinformatics 25(14):1754–1760. https://doi.org/10.1093/bioinformatics/btp324
Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, Marth G, Abecasis G, Durbin R (2009) The sequence alignment/map format and SAMtools. Bioinformatics 25(16):2078–2079. https://doi.org/10.1093/bioinformatics/btp352
Luan G, Bao G, Lin Z, Li Y, Chen Z, Li Y, Cai Z (2015) Comparative genome analysis of a thermotolerant *Escherichia coli* obtained by genome replication engineering assisted continuous evolution (GREACE) and its parent strain provides new understanding of microbial heat tolerance. N Biotechnol 32(6):732–738. https://doi.org/10.1016/j.nbt.2015.01.013
MacMillan SV, Alexander DA, Culham DE, Kunte HJ, Marshall EV, Rochon D, Wood JM (1999) The ion coupling and organic substrate specificities of osmoregulatory transporter ProP in *Escherichia coli*. Biochim Biophys Acta 1420(1–2):30–44. https://doi.org/10.1016/S0005-2736(99)00085-1
Martin M (2011) Cutadapt removes adapter sequences from high-throughput sequencing reads. EMBnet J 17:10–12. https://doi.org/10.14806/ej.17.1.200
Martinez A, Grabar TB, Shanmugam KT, Yomano LP, York SW, Ingram LO (2007) Low salt medium for lactate and ethanol production by recombinant *Escherichia coli* B. Biotechnol Lett 29(3):397–404. https://doi.org/10.1007/s10529-006-9252-y
Matsushita K, Azuma Y, Kosaka T, Yakushi T, Hoshida H, Akada R, Yamada M (2016) Genomic analyses of thermotolerant microorganisms used for high-temperature fermentations. Biosci Biotechnol Biochem 80(4):655–668. https://doi.org/10.1080/09168451.2015.1104235
Mehta R, Singhal P, Singh H, Damle D, Sharma AK (2016) Insight into thermophiles and their wide-spectrum applications. 3 Biotech 6(1):81. https://doi.org/10.1007/s13205-016-0368-z
Miller EM, Nickoloff JA (1995) *Escherichia coli* electrotransformation. Methods Mol Biol 47:105–113. https://doi.org/10.1385/0-89603-310-4:105
Morales GP, Delaye L (2020) Was LUCA a hyperthermophilic prokaryote? The impact-bottleneck hypothesis revisited. In: Souza V, Segura A, Foster J (eds) Astrobiology and cuatro ciénegas basin as an analog of early earth. Cuatro ciénegas basin: an endangered hyperdiverse oasis. Springer, Cham. https://doi.org/10.1007/978-3-030-46087-7_3
Mordukhova EA, Pan JG (2013) Evolved cobalamin-independent methionine synthase (*MetE*) improves the acetate and thermal tolerance of *Escherichia coli*. Appl Environ Microbiol 79(24):7905–7915. https://doi.org/10.1128/AEM.01952-13
Mordukhova EA, Lee HS, Pan JG (2008) Improved thermostability and acetic acid tolerance of *Escherichia coli* via directed evolution of homoserine o-succinyltransferase. Appl Environ Microbiol 74(24):7660–7668. https://doi.org/10.1128/AEM.00654-08
Pérez-Morales G, Caspeta L, Merino E, Cevallos MA, Gosset G, Martinez A (2024) Simultaneous saccharification and fermentation for -lactic acid production using a metabolically engineered *Escherichia coli* adapted to high temperature. Biotechnol Biofuels Bioprod 17:132. https://doi.org/10.1186/s13068-024-02579-1
Pérez-Morales G, Martínez-Conde KV, Caspeta L, Merino E, Cevallos MA, Gosset G, Martinez A (2025) Thermally adapted *Escherichia coli* keeps transcriptomic response during temperature upshift exposure. Appl Microbiol Biotechnol 109:120. https://doi.org/10.1007/s00253-025-13495-1
Potrykus K, Cashel M (2008) (p)ppGpp: still magical? Annu Rev Microbiol 62:35–51. https://doi.org/10.1146/annurev.micro.62.081307.162903
Quan J, Tian J (2011) Circular polymerase extension cloning for high-throughput cloning of complex and combinatorial DNA libraries. Nat Protoc 6(2):242–251. https://doi.org/10.1038/nprot.2010.181
Reyes-Gonzáles LT, Molina-Vázquez ER, Brito-Alarcón E, Peralta-Gil M, Martínez-Jiménez A (2021) Implementación del Sistema CRISPR/Cas9 en *Escherichia coli* y sus aplicaciones en la ingeniería de vías metabólicas. Revista de la Alta Tecnología y Sociedad. 13(1). (In Spanish). https://www.academiajournals.com/s/H320ReyesGonzales-ATS-V13N1-2021-42-51.pdf. Accessed 15 Jan 2025
Riehle MM, Bennett AF, Long AD (2001) Genetic architecture of thermal adaptation in *Escherichia coli*. Proc Natl Acad Sci U S A 98(2):525–530. https://doi.org/10.1073/pnas.98.2.525
Rodríguez-Verdugo A, Tenaillon O, Gaut BS (2016) First-step mutations during adaptation restore the expression of hundreds of genes. Mol Biol Evol 33(1):25–39. https://doi.org/10.1093/molbev/msv228
Rudolph B, Gebendorfer KM, Buchner J, Winter J (2010) Evolution of *Escherichia coli* for growth at high temperatures. J Biol Chem 285(25):19029–19034. https://doi.org/10.1074/jbc.M110.103374
Sandberg TE, Pedersen M, Lacroix RA, Ebrahim A, Bonde M, Herrgard MJ, Palsson BO, Sommer M, Feist AM (2014) Evolution of *Escherichia coli* to 42 °C and subsequent genetic engineering reveals adaptive mechanisms and novel mutations. Mol Biol Evol 31(10):2647–2662. https://doi.org/10.1093/molbev/msu209
Sandberg TE, Salazar MJ, Weng LL, Palsson BO, Feist AM (2019) The emergence of adaptive laboratory evolution as an efficient tool for biological discovery and industrial biotechnology. Metab Eng 56:1–16. https://doi.org/10.1016/j.ymben.2019.08.004
Seemann T (2014) Prokka: rapid prokaryotic genome annotation. Bioinformatics 30(14):2068–2069. https://doi.org/10.1093/bioinformatics/btu153
Tenaillon O, Rodríguez-Verdugo A, Gaut RL, McDonald P, Bennett AF, Long AD, Gaut BS (2012) The molecular diversity of adaptive convergence. Science 335(6067):457–461. https://doi.org/10.1126/science.1212986
Utrilla J, Licona-Cassani C, Marcellin E, Gosset G, Nielsen LK, Martinez A (2012) Engineering and adaptive evolution of *Escherichia coli* for *d*-lactate fermentation reveals GatC as a xylose transporter. Metab Eng 14(5):469–476. https://doi.org/10.1016/j.ymben.2012.07.007
Utrilla J, Vargas-Tah A, Trujillo-Martínez B, Gosset G, Martinez A (2016) Production of *d*-lactate from sugarcane bagasse and corn stover hydrolysates using metabolic engineered *Escherichia coli* strains. Bioresour Technol 220:208–214. https://doi.org/10.1016/j.biortech.2016.08.067
Vavitsas K, Glekas PD, Hatzinikolaou DG (2022) Synthetic biology of thermophiles: taking bioengineering to the extremes? Appl Microbiol 2(1):165–174. https://doi.org/10.3390/applmicrobiol2010011
Wang X, He J, Le K (2018) Making point mutations in *Escherichia coli* BL21 genome using the CRISPR-Cas9 system. FEMS Microbiol. Lett. 365(14). https://doi.org/10.1093/femsle/fny060
Yang D, Prabowo CPS, Eun H, Park SY, Cho IJ, Jiao S, Lee SY (2021) *Escherichia coli* as a platform microbial host for systems metabolic engineering. Essays Biochem 65(2):225–246. https://doi.org/10.1042/EBC20200172
Ying BW, Matsumoto Y, Kitahara K, Suzuki S, Ono N, Furusawa C, Kishimoto T, Yomo T (2015) Bacterial transcriptome reorganization in thermal adaptive evolution. BMC Genomics 16:802. https://doi.org/10.1186/s12864-015-1999-x
Ziegler CA, Freddolino L (2023) Escherichia coli leucine-responsive regulatory protein bridges DNA in vivo and tunably dissociates in the presence of exogenous leucine. mBio 14(2):e0269022. https://doi.org/10.1128/mbio.02690-22

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